

SANJAY S

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RESEARCH INTERESTS

- Superconducting quantum hardware — transmon and fluxonium qubit design
- Quantum processor architecture and scalable multi-qubit systems
- Microwave resonators and circuit quantum electrodynamics
- Inverse design, electromagnetic simulation, and quantum control electronics

SUMMARY

- Final-year Electronics and Communication Engineering student working on superconducting quantum hardware, focused on turning quantum-device specifications into fabrication-ready designs — from transmon circuits and readout resonators to multi-qubit architectures — using circuit modelling, electromagnetic simulation, and optimization in Qiskit Metal, scQubits, KQCircuits, QuTiP, Ansys HFSS, Ansys Q3D, and Keysight ADS QuantumPro.
- Research Intern at C-DAC Bengaluru, designing and verifying superconducting qubits, developing RF data-converter IP for quantum control electronics, and working with photonic laboratory systems including DPS-QKD setups and photon-coincidence measurement.
- Finalist in the VLSID Design Contest 2026 for a 4-qubit flip-chip transmon processor and in the RIT Quant-A-Thon 2026 for a dispersive shift (χ) inverse-design tool; additional experience in FPGA and VLSI design, RF electronics, and applied machine learning.

EDUCATION

Chennai Institute of Technology and Applied Research, Chennai

Aug 2023 – Present

B.E. Electronics and Communication Engineering | CGPA: 9.16

Cholan Matriculation Higher Secondary School, Kanchipuram

2019 – 2022

Grade 12: 92% (2022) | Grade 10: 98% (2020)

WORK EXPERIENCE

Research Intern | Quantum Technology Group, C-DAC Bengaluru

Jan 2026 – Aug 2026

- Designed and simulated superconducting qubits using Qiskit Metal, scQubits, KQCircuits, Ansys HFSS, Ansys Q3D, and Keysight ADS QuantumPro for quantum processor development.
- Developed RF ADC, RF DAC, and RF data-converter IP on a Zynq UltraScale+ RFSoc, and studied NCO and AWG signal-generation techniques for quantum control electronics.
- Worked with quantum photonic laboratory systems including DPS-QKD setups — lasers, RF drivers and attenuators, optical couplers, polarization beam splitters, photon detectors, and digital storage oscilloscopes; performed photonic coincidence experiments.

Project Intern, RNG Randomness Quality Detection (ML) | SparQ Internship Program, QNu Labs

May 2026 – Jul 2026

- Trained and tuned the five-model tree-based ensemble (CatBoost, XGBoost, LightGBM, Random Forest, Extra Trees) at the core of RSAF — a four-person team framework attributing binary randomness streams to algorithmic (PRNG) or physical (TRNG/QRNG) sources — classifying a 64-dimensional statistical feature set spanning entropy, spectral, correlation, and complexity descriptors.
- Benchmarked the ensemble against every individual base learner across accuracy, precision, recall, and F1-score, reaching **74.3% accuracy** and a **73.2% F1-score** — ahead of all five base learners.
- Evaluated across 37 generators (~4.15 GB) including QNu Labs' proprietary photonic QRNG output; the full team pipeline reached **81.21%** mean generator-level attribution confidence.

VLSI & FPGA Design Engineer (R&D) | ULOG3, Chennai

Oct 2023 – Dec 2025

- Contributed front-end RTL design and synthesis (Cadence Genus) to OLI, an on-chip-learning neuromorphic processor, validated on a ZCU102 FPGA.
- Designed and launched the ULOG-3 CubeSat to the stratosphere (Sep 2024), covering circuit design, power management, and payload integration.

PROJECTS

4-Qubit Flip-Chip Transmon Processor

VLSID Design Contest 2026 — Finalist

- Designed a Q-chip / C-chip flip-chip architecture in Keysight ADS QuantumPro, physically separating the qubit layer from control and readout routing to cut routing congestion and electromagnetic crosstalk.
- Verified the design across eigenmode, full-EM, and energy-participation (EPR) analyses, obtaining a **simulated T_1 of 3.52–4.61 μ s** across Q1–Q4.
- Exported a fabrication-ready GDS layout from the verified geometry in KLayout.

RSAF — Randomness Source Attribution Framework

QNu Labs, SparQ Internship — team project

- Built and tuned the five-model calibrated ensemble over a 64-feature statistical fingerprinting pipeline (entropy, spectral, correlation, and complexity descriptors) that distinguishes algorithmic (PRNG) from physical (TRNG/QRNG) randomness sources.
- Four-person team project: the team's isotonic probability calibration and hierarchical window $\rightarrow$ file $\rightarrow$ generator voting turned raw classifier scores into a calibrated attribution confidence, improving the Brier score from 0.21 to **0.16**.

Dispersive Shift (χ) Optimizer for Superconducting Qubit Design

RIT Quant-A-Thon 2026 — Finalist

- Built a closed-form inverse-design engine converting a target qubit frequency, resonator frequency, and dispersive shift (f_q, f_r, χ) directly into a complete, fabricable transmon and readout-resonator geometry.
- Implemented multi-fidelity Bayesian optimization (Optuna TPE) across an analytic $\rightarrow$ Q3D/MoM $\rightarrow$ HFSS+EPR fidelity ladder, reserving full electromagnetic solves for verification to keep the design loop fast.
- Developed the Streamlit design GUI and cross-checked solver output against Keysight ADS QuantumPro reference values. Represented Team Transmon Titans as a finalist in the dispersive-shift optimization track.

FPGA-Based Quantum Control Electronics: RF Data Converter

C-DAC Bengaluru — 2026

- Configured the RF Data Converter IP on a ZCU111 RFSoc — RF-ADC Tile_224 and RF-DAC Tile_229 at **1.47456 GSPS**, mixer bypassed, 1 $\times$ decimation, on a 184.32 MHz AXI4-Stream fabric clock.
- Simulated the AXI4-Lite power-on register sequence and closed synthesis and implementation for the xczu28dr-ffvg1517-2-e. (*Vendor reference block design — configured, simulated, and implemented rather than authored.*)

GDS Layer Conventions and Cross-Tool Mapping

C-DAC Bengaluru — 2026

- Mapped layouts out of KQCircuits, Qiskit Metal, and Keysight ADS into the ConScience Q-MPW foundry stack by function rather than by layer number: the base layer inverts between tools — KQCircuits and ConScience draw the gap where Qiskit Metal and Keysight ADS draw the metal — so renumbering alone would swap metal and gap across the whole chip.

TECHNICAL SKILLS

Quantum Device Design: Qiskit Metal, scQubits, KQCircuits, QuTiP, Cirq, Qiskit

Electromagnetic & RF Simulation: Ansys HFSS, Ansys Q3D, Keysight ADS, Keysight QuantumPro

Hardware & Control Electronics: RTL / Verilog, Xilinx Vivado, Cadence Genus, FPGA IP development for RF ADC, RF DAC, and RF data converters

Machine Learning & Optimization: CatBoost, XGBoost, LightGBM, scikit-learn, SHAP, Optuna, Streamlit

Layout & Fabrication Tooling: KLayout, GDS export and cross-tool layer mapping

Programming & Platforms: Python, C, Verilog, Linux, Windows

ACHIEVEMENTS

- Winner, MSME Idea Hackathon 4.0** — ₹15 Lakh award for a neuromorphic EO payload small-satellite concept.
- Finalist, VLSID Design Contest 2026** — 4-qubit flip-chip superconducting transmon processor.
- Finalist, RIT Quant-A-Thon 2026** — dispersive shift (χ) optimization for superconducting qubit design, Team Transmon Titans.
- Finalist, Amaravati Quantum Valley Hackathon 2025** — quantum machine learning solution for fraud detection.
- ULOG-3 CubeSat** — designed, built, and launched to the stratosphere, Sep 2024.

COURSEWORK, WORKSHOPS & SEMINARS

Courses: The Hardware of a Quantum Computer (TU Delft) · Fundamentals of Superconducting Qubits (QuTech Academy) · Quantum Computing with Semiconductor Technology, Professional Certificate (TU Delft) · Introductory Course on Quantum Computing (IISER Pune) · Introduction to Quantum Computing (Qubit by Qubit / The Coding School) · Quantum Algorithms and Qiskit (IIT Madras)

Workshops: Quantum Superconducting Circuit Design Program (Womanium Quantum) · Quantum Technology / Semiconductor Device / Nanotechnology Simulation and Experimental Characterization (Impulse Technology) · Qiskit Fall Fest 2024 (IBM Quantum) · Demystifying Quantum Computing (IAPT) · Short Term Training Program on Quantum Computing

Seminars: Novel Superconducting Qubits (Google Quantum AI) · Qiskit Global Summer School 2024 (IBM Quantum) · Quantum Artificial Intelligence for Renewable and Sustainable Energy